Group Coursework: Forecasting Oil Prices

Autumn 2025 Financial Econometrics

Imperial College Business School
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- **Written Report (2,000 words):** Wednesday Week 7 (report and program files) – 20%
- **Presentation (5 minutes plus Q &A):** Week 8 (in class) – 5%

1. Coursework Overview

Imagine you are a junior analyst working for an oil trading desk. Your task is to develop **a forecasting model** that can help a senior trader form expectations about oil futures returns and identify potential trading opportunities.

The trader is not an econometrician. They care about whether your model produces useful, **interpretable signals** that can inform trading decisions. They also want to know what data and reasoning underpin your forecasts, how reliable the model is, and what risks or limitations they should be aware of.

Your role is therefore twofold:

1. Technical – Apply econometric methods to forecast oil futures, carefully selecting the dependent variable (e.g. CL1 price, CL1 returns, or the CL2–CL1 spread), the relevant predictors (from the macroeconomic and industry dataset provided), and the appropriate data frequency. Evaluate your model with diagnostic tests and out-of-sample performance checks.
2. Communicative – Translate your results into clear economic and trading insights that a busy oil trader can understand and act on.

Deliverables:

- A written report (2,000 words) that balances econometric rigour with trader-oriented communication.
- A 5-minute presentation plus Q & A that highlights only the most important results and implications for trading.

2. Data

The dataset (*Oil Data.xlsx*) contains:

- CL1: front-month crude oil futures price.
- CL2: second-month crude oil futures price.
- Additional macroeconomic and industry data (e.g. exchange rates, inventories, interest rates, global demand proxies).

It is your task to decide which variables are relevant for forecasting. Be explicit about how you deal with differences in frequency across variables.

3. Methodology

You are free to select your approach, provided you justify your choice. Techniques introduced in class include:

- Predictive regressions using OLS.
- ARIMA and ARIMAX models.
- LASSO and Ridge Regressions.

Some groups may be familiar with more advanced statistical techniques and keen to show them. you may use machine learning methods (e.g. random forests, neural networks), **BUT**: They are not the focus of this assignment! If you choose to include them, show how their results compare to more traditional econometric approaches. You will not be rewarded simply for using advanced tools — the emphasis is on justification, transparency, and communication.

You are free to use any appropriate econometric software. The module and tutorials mostly cover R implementations, but you are welcome to use other software, such as for example Python, Julia, Matlab, Eviews, Stata, or other statistical software you feel comfortable with.

4. Grading Criteria

Your grade will depend on:

- **Clarity of explanation** – Could a trader understand your reasoning?
- **Diagnostics** – Did you check whether your model satisfies basic econometric assumptions?
- **Interpretation** – Can you explain why the model performs as it does and what it means economically?

Your report should:

- Justify your dependent variable, data frequency, and choice of predictors.
- Present diagnostics (stationarity, autocorrelation, heteroskedasticity, parameter stability).
- Evaluate forecast performance using out of sample metrics. Your model evaluation must include comparison with at least one simple naïve benchmark.
- Translate results into economic and trading insights, possibly outlining a simple trading rule.

Report

A strong report will:

- Be clearly structured, with abstract, introduction, methodology, results, and conclusion.
- Frame the research question in **trader-relevant terms** ("How can we forecast oil returns to support trading decisions?").
- Clearly explain data choices and frequency alignment.
- Justify the econometric methodology and report diagnostic checks.
- Present results in a clear, visual, and intuitive way.
- Offer thoughtful interpretation: *What do these results mean for an oil trader?*
- Be accessible to a finance professional with limited econometrics background.

Presentation

An effective presentation will:

- Be pitched as a **short analyst briefing to a trading desk**.
- Highlight only the most important results and trading implications (not every detail from the report).
- Use simple, persuasive visuals.
- Avoid technical jargon — focus on what the trader needs to know.
- I would suggest at most two presenters, with the rest of the team prepared for Q&A.

5. Hints and Guidance

- **Target Variable** – Will you forecast the level of CL1, its returns, or the CL2–CL1 spread? Which is most useful for a trader? What econometric properties do they have?
- **Frequency** – Would a trader care more about daily, weekly, or monthly forecasts? What are the pros and cons of each choice? Make sure to stick with one frequency once you have made your choice.
- **Predictors** – How do macroeconomic and industry variables help? Be selective and explain why you include or exclude them.
- **Depth vs. Breadth –** Avoid the trap of trying to do too much. After some initial exploration, hone in on one or two models that you want to estimate and focus your energy on explaining those clearly and convincingly to your target audience.
- **Model Comparison** – Use appropriate metrics to compare models. Out-of-sample performance matters more than in-sample fit.
- **Model Complexity** – A more complex model is not automatically better. You will be rewarded for selecting the right level of complexity for the problem, justifying your choice, and communicating the trade-off between interpretability and predictive power. A simple, well-motivated model explained clearly is often more valuable to a trader than a black-box approach.
- **Trading Angle** – If your forecasts show promise, outline how they could translate into a trading rule (while noting practical limitations).
- **Assessment Focus** – You are NOT graded on whether your forecasts "beat the market." You are graded on how well you justify your model, test it, and communicate its relevance to trading.